

Hierarchical Optimization Technique for Placement of Visual Landmarks for Robot Navigation in Repetitive Structured Indoor Environments

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Abstract. Autonomous indoor navigation poses significant challenges, particularly in visually repetitive GPS-denied environments, such as hotel corridors, hospital rooms, and shopping malls. Visual markers like ArUco (Augmented Reality University of Córdoba) tags can serve as reliable navigation beacons by providing unique landmarks for robot localization. However, determining the optimal placement of these markers is crucial, as excessive placement creates visual clutter while insufficient placement makes reliable robot navigation challenging. To overcome this problem, the proposed work aims to develop a hierarchical optimization technique that automates the placement of visual markers in the given environment, enabling autonomous navigation. Three different types of markers, namely functional, junction, and corridor tags, are used to provide structural information to the robot. Mathematical feasibility analysis is carried out to select optimal marker parameters based on camera intrinsics and environmental constraints. Greedy search-based hierarchical placement logic is used to determine the optimal placement and density of markers. Simulation study results in marker availability of 94.2%, while reducing the marker usage by 82% in structured environment. A reliable 100% marker availability is observed at junctions, where robot navigation is challenging. A mean visibility of 1.011 tags per frame is reported, thereby offering autonomous indoor navigation more accessible for real-world applications. Hardware validation is also carried out, which demonstrated the reliability of the proposed technique.

Keywords: Autonomous Indoor Navigation, ArUco Markers, Greedy Optimization, Vision-Based Localization, Mobile Robotics, Navigation Beacons.

1 Introduction

Indoor positioning systems(IPS) have been widely explored by many researcher[1], which enables the mobile robot to complete tasks like pick and place, patrol, and

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elevators operations [2], The high accuracy of IPS is one of the primary requirement to ensure that the mobile robot can complete the task and ensure the safety of the human-machine environment. This demand to achieve a centimetre-level positioning accuracy of mobile robots, which motivates researchers to continue to explore the positioning of indoor mobile robots [3]. Researchers have [5,6] explored the application of Simultaneous Localization and Mapping (SLAM) to address localization in unknown environment. However, repetitive and structural regularity in the environment hinders the SLAM from detecting unique landmarks makes it inefficient.

Navigation beacons, which emerge as alternatives enabling accurate localization for indoor navigation in such structured repetitive environments like hotel corridors, hospitals, shopping malls and other larger public places. visible markers are predefined because of their simple 2D patterns, easier preparation and placements. They are designed to be recognized by vision systems. ArUco, April Tags and QR codes are the popular types of visible markers that serve as ground truth for IPS. ArUco Markers are synthetic square binary markers based on a simplified internal matrix. It is specifically designed for real-time performance and low computational overhead, allowing for rapid detection even on resource-constrained hardware [[9]. They carry a smaller data payload compared to QR codes, though this is negligible for simple ID-based localization. Hence in this work, ArUco markers are selected over QR and April Tags due to their superior balance of detection speed and localization reliability. To Navigate and localise an autonomous mobile robot in corridor-like structured indoor environments, visual markers are placed in multiple locations.

Literature investigating autonomous navigation of mobile robots[10,11,12] using visual landmarks and text detection is widely reported. Optimal placement of these visual landmarks is one of the prime factors that determines the reliability of autonomous navigation. Several techniques for the placement and selection of visual markers that enable accurate navigation of mobile robots have been reported. Placement of visual beacons can compensate for sparse environment knowledge [13] and enable robots to maintain an accurate trajectory. Optimal placement of ArUco tags provides accurate pose estimation in robots [14,15]. Perception-aware tag placement ensures robust indoor localisation for Unmanned Aerial Vehicles (UAVs) by accounting for the robot's specific field of view and perception limitations [16]. Placement of tags requires extensive knowledge of robot perception and environmental characteristics, making the process challenging and often requiring expert intervention.

To address these challenges, a visibility-aware hierarchical optimisation framework is proposed for tag placement. The work aims to automate tag placement by gathering initial inferences from the manual observations, which eliminates the need for expert intervention. It also categorises beacons based on environmental priority, which provides structural knowledge for robot navigation. This approach uses mathematical feasibility analysis and employs a greedy selection algorithm to maximise the robot's lateral field of view and minimise blackout in repetitive, structured layouts.

Major contribution of the proposed work is as follows,

- Calibration of the onboard camera system to determine the size and ensure accurate ArUco markers
- Configuration of functionally specific markers and feasibility analysis for marker placement
- Design of a hierarchical optimization technique for the placement of ArUco markers by analyzing spatial constraints, visibility requirements, and navigation performance.
- Validation of the proposed technique using simulation and in a real-time environment.

The paper is organized as follows: Section 2 presents the environmental setup, and Section 3 details the methodology implemented. Section 4 presents the results and discusses the findings, limitations, and future work. Section 6 concludes the paper, summarizing the key contributions and implications of the research.

2 Environmental Setup

A hotel-like indoor corridor environment characterized by repetitive structures is designed for the proposed study, as shown in Figure 1a. The environment consists of functional areas such as doors and junctional zones, which are error-prone zones and have a higher possibility for localization errors. Constraints are defined based on the geometry of the corridor to determine the lateral visibility and the required bearing angles for pre-turn and post-turn junction availability. Identical simulated (1b) and real-time environments (1c) are created for performance evaluation.



Fig. 1: Details of the experimental environment considered in this study

The optimized coordinates are imported into the 3D simulation environment. Figure 1b shows the Webots interface with the setup environment. The robot's

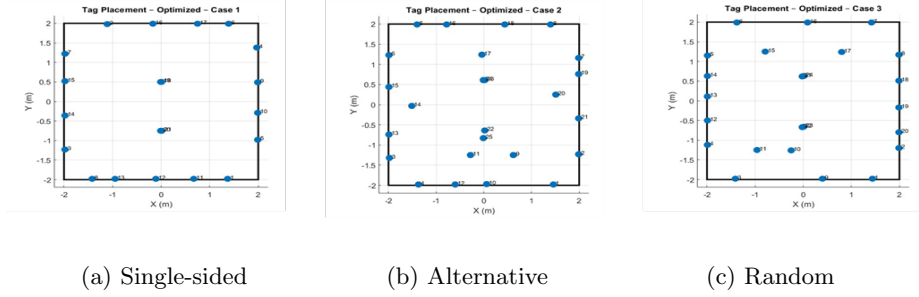


Fig. 2: Manual placement of the marker under various constraints

performance is monitored in real-time to verify that the tags placed satisfy the navigation requirements in dynamic settings. The validation uses the Yahboom Jetbot (Figure 1c), with a scaled-down environment as shown in Figures 1a and 1b, an AI-driven smart robot based on the NVIDIA Jetson Nano 4GB (B01). The Jetson Nano enables real-time execution of the ArUco detection and localization stack. The robot uses Yahboom IMX219 HD Camera, featuring an 8MP IMX219 chip with a maximum resolution of 3280×2464 . 160-degree field of View and 3.15mm focal length, and 2.35 aperture with 14.3% distortion

3 Methodology

Figure 3 shows the proposed hierarchical optimization technique to maximize tag visibility, which ensures robot localization reliability within constrained indoor environments. Initially, feasibility analysis using camera intrinsics and corridor geometry is carried out, which results in a robot perception model. Visibility matrix to map candidate tag locations against all the sampled robot poses is generated. An optimal tiered strategy is employed that prioritizes high-value areas, specifically targeting functional zones like doorways and critical junctions where manoeuvring requires high tag density. Other locations are addressed using a greedy selection algorithm that balances coverage and redundancy against the infrastructure costs.

3.1 Inference Generation from Manual Placement

The beacons are categorized into three types based on placement, i.e. Functional Tags, which are placed at doors or task areas, Junction Tags, which are placed at intersections, and Corridor Tags, which are for straight-line navigation.

Visibility constraints are derived using the environment and camera parameters, which are used as baseline constraints for the manual placement technique. The cases are analyzed under optimal and non-optimal configurations. Figure 4 shows the methodology for manual placement. It is observed in non-optimal cases that tags are placed uniformly with an interval of 25cm both in single side

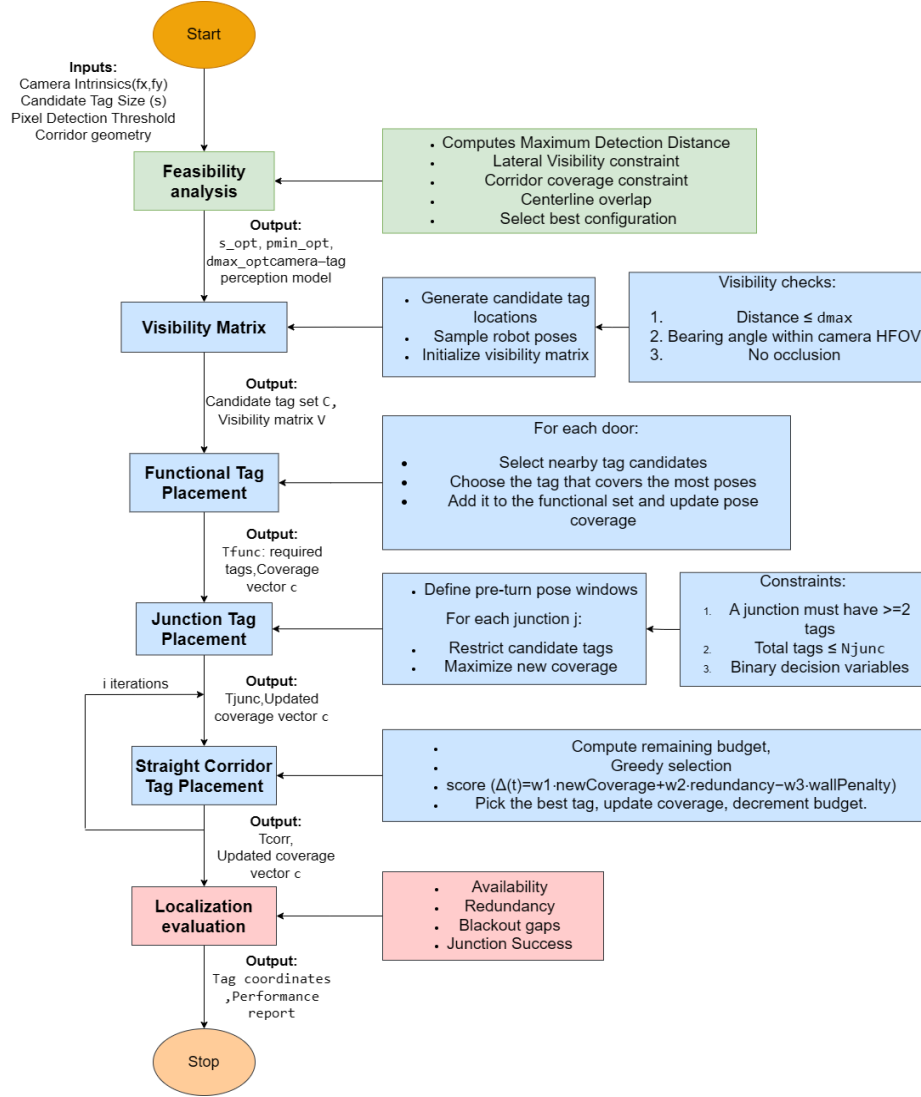


Fig. 3: Flow chart of the proposed hierarchical optimization technique for visual landmark placement

and alternating sides of the corridor walls. On the other hand, optimal cases enable tag deployment subjected to specific visibility constraints.

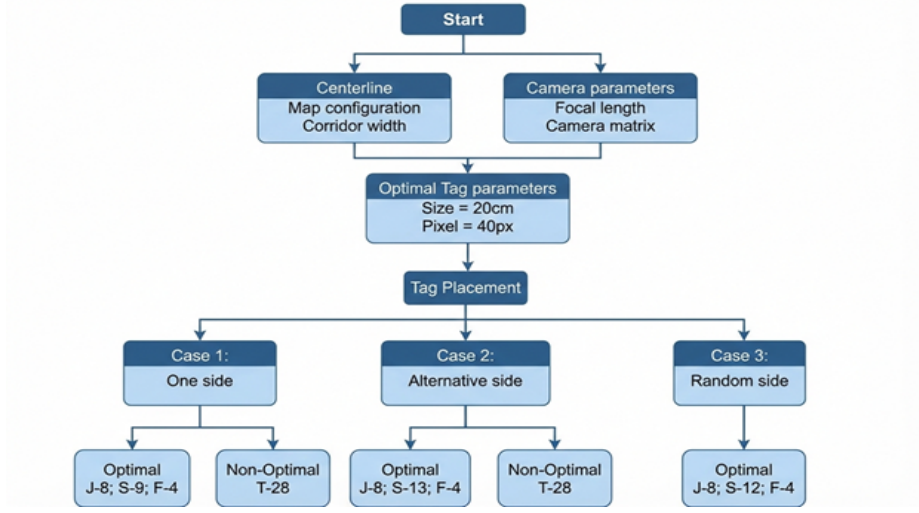


Fig. 4: Inference generation from manual tag placement

The inferences noted from manual placements are sensitive to tag dimensions(s). For instance, smaller tags provide insufficient localisation, while large tags create over-dense tags, resulting in a cluttered camera field of view (FOV). Therefore, an optimal tag size that suits the camera requirements and the environment is analysed. Uniform placement yields high redundancy, which is useful for obscured tags but inefficient for infrastructure requirements. Junction and functional tags are mandatory for deterministic navigation. Pre- and Post-junction visibility is critical for accurate localization and reduction of blind spots. Alternating placement is more effective for wide corridors, whereas single-side placement is sufficient for narrow paths. These are some of the inferences generated from manual placement, which serve as a knowledge base for the optimization framework.

3.2 Hierarchical Optimization Framework

The objective is to select a minimal subset of ArUco tags S from a candidate pool C , such that the robot's path P achieves maximum localization availability while maintaining lower redundancy. The following are the constraints used,

1. Detection feasibility - A tag is only selected if its distance d satisfies as shown in eq.1

$$\frac{f_s}{P_{x,\max}} \leq d \leq \frac{f_s}{P_{x,\min}} \quad (1)$$

2. Geometric visibility – The bearing angle relative to the robot must satisfy the constraint described in eq.2.

$$\text{Angle} \leq \frac{\text{HFOV}}{2} \quad (2)$$

3. Mandatory placement – Functional and junction tags to be selected, as shown in condition eq.3

$$x_j = 1 \quad \text{if } A_j \in F \cup J \quad (3)$$

A multi-stage optimization algorithm as shown in Figure 3 is designed to optimize the placement and number of ArUco tags used to ensure reliable indoor robot navigation with minimal infrastructural changes. It refers to the trade-off between the reduction of Aruco tags while increasing the localization availability. The algorithm utilizes camera intrinsics (f_x, f_y) and pixel thresholds (PxThr) to calculate the maximum detection distance ($d_{\max}$), as shown in eq. 4. A visibility matrix(V) is derived by sampling potential robot poses along the centerline of the robot path and evaluated for distance, bearing angle and occlusion constraints.

$$D_{\max} = \frac{f \cdot s}{p_{\min}} \quad (4)$$

A configuration is feasible only if it satisfies two environmental constraints as follows,

1. Lateral visibility: $d_{\max}$ must exceed the maximum wall offset to ensure the robot can visualize the side walls from the corridor centerline
2. Corridor coverage: $d_{\max}$ must exceed the maximum corridor width to ensure continuous coverage.

Candidate tags (C_j) are generated along all wall surfaces with a uniform spacing, enabling the set of tags to satisfy the visibility constraints, as shown in eq. 5, It is described as binary visibility state for every given robot pose (P_i).

$$V_{i,j} = 1 \text{ if } (dist \leq d_{\max}) \cap (|bearing| \leq \frac{\text{HFOV}}{2}) \cap (n \cdot v > 0) \quad (5)$$

3.3 Multi-Stage Tag Placement

The algorithm is divided into three sequential stages for tag placement,

Stage 1 Functional Tags

The algorithm initially prioritizes high-priority task areas. Candidate tags that are closest to the doorways are selected to ensure task-specific localization.

Stage 2 Junction Tags

Tags are selected for placement at junctions in the corridor, where pre-turn and post-turn of robot leads to localization errors and reliable markers are required. In our work, L- and T- based intersections are considered as junctions.

Stage 3 Corridor Tags

The remaining straight corridors are filled using a greedy iterative algorithm that selects tags from the candidate pool based on a weighted objective function as shown in eq. 6

$$\text{Score} = \text{NewCoverage} - \text{Penalty} \quad (6)$$

As straight corridors are less error-prone, a penalty is applied if a candidate is placed on the same side of the selected location. It enables the selection of candidate tags that minimize the blackouts while avoiding redundancy.

Overall, each stage of the algorithm contains its own optimization goal, i.e functional tags are required for optimization of task points and improvement in localization. Junction tags improve navigation safety with the utilization of mandatory pre and post turn requirements. The straight corridor stage reduces the gap while increasing the total localization availability of the markers for the robot.

4 Results and Discussion

The proposed technique is evaluated in the structured environment as described in Section 2. Initially, tags are placed manually at a uniform distance of 25cm for both single and alternate wall constraints. Experimental analysis reports a high redundancy, but at the same time resulted in higher availability as shown in Table 1. The three optimal cases, namely, single side(C1), alternate side(C2), and random side(C3), are also evaluated under optimal and non-optimal conditions. It is observed that optimal C3 achieved the highest performance with 89% overall availability and 75% availability at junctions as shown in Figure ?? . It also maintained a low redundancy(11%) and the smallest maximum visibility gap of 0.5m.

Table 1: Performance evaluation of manual tag placement

Case	Overall Availability (%)	Junction Availability (%)	Redundancy (%)	Missed Junction
Optimal C1	75	56.25	11.11	7
Optimal C2	86.11	68.75	13.88	5
Optimal C3	88.88	75	11.11	4
NonOpt C1	77.77	68.75	44.44	2
NonOpt C2	77.77	68.75	30.55	2

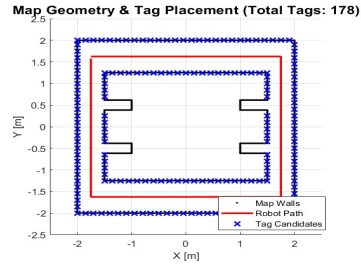
Optimal tag size for the given environment is also determined in the proposed work. The pixel threshold increases as the maximum detection distance decreases for ArUco tags. Hence, ArUco markers of different sizes (0.10m, 0.15m, and 0.20 m) are evaluated as in Table 2. The proposed feasibility analysis technique selects an optimal configuration of 0.2m tags with a 40px threshold, providing a maximum detection distance (dmax) of 1.1m. (Ref Table 2)

A set of candidate tag locations is determined as shown in Figure 5a. The algorithm prioritizes door and other task-specific zones, where robot localization is critical. It is observed that each doors are allotted with functional tags (indicated by blue color dots) in the given environment, as in Figure 5b, and leading to a 97.8% reduction from candidates' tags while achieving an overall reliable localization availability. Junction tags were added to ensure accurate navigation at critical turns. Junction tags are added at each corner of the environment as in

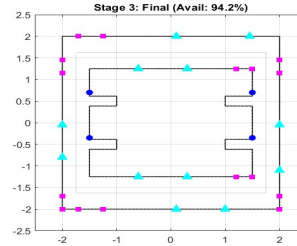
Table 2: Feasibility analysis for determination of tag dimension

Tag Size (m)	Pixel Threshold	Maximum Distance (m)	Detection Status
0.1	30	0.73	Fail: Width
0.1	40	0.55	Fail: Width
0.1	50	0.44	Fail: Width
0.15	30	1.10	OK
0.15	40	0.82	Fail: Width
0.15	50	0.66	Fail: Width
0.2	30	1.46	OK
0.2	40	1.10	OK
0.2	50	0.88	Fail: Width

5b, indicated by pink colour dots. It increases the availability by 60% and significantly reduces the blackout length to 2.3m. Finally, the straight tags (indicated by cyan color dots) for the remaining locations are placed using a greedy heuristic algorithm. Over the 12 iterations, the system evaluated the candidate pool and reached a stable coverage point. The final configuration used 32 tags, which gave an 82% reduction in tag usage to achieve 94.2% localization availability as in Table 3. Optimal tag placements also provide a minimal blackout length of 0.3m.



(a) Possible candidates for placement



(b) Finalised placement

Fig. 5: Resultant optimized placement of tags

Table 3: Details of optimized tag placement

Type of Tag	Tags Used (%)	No. Tags Used	Total Candidates	Reduction (%)	Localization Availability (%)	Total Blackout Length (m)
Functional	2.2	4	178	97.8	21.7	4.9
Junction	11.2	20	178	88.8	60.1	2.3
Straight	18	32	178	82.0	94.2	0.3

4.1 Performance Evaluation in Simulation Environment

The performance of the proposed technique is evaluated using Webots R2025a robot simulator. The resultant optimized tag coordinates are imported, and visual markers are placed in the respective location. A mobile robot with a camera is set to navigate the simulated environment. The camera feed is acquired and analysed to evaluate the tag visibility as in ???. Tag visibility over the distance is analysed (Ref 6b) to evaluate the redundancy and blackout. Blackout rate is reduced to 6.32 % (Ref 4) at significantly smaller intervals(Figure 6c, which results in reliable navigation.

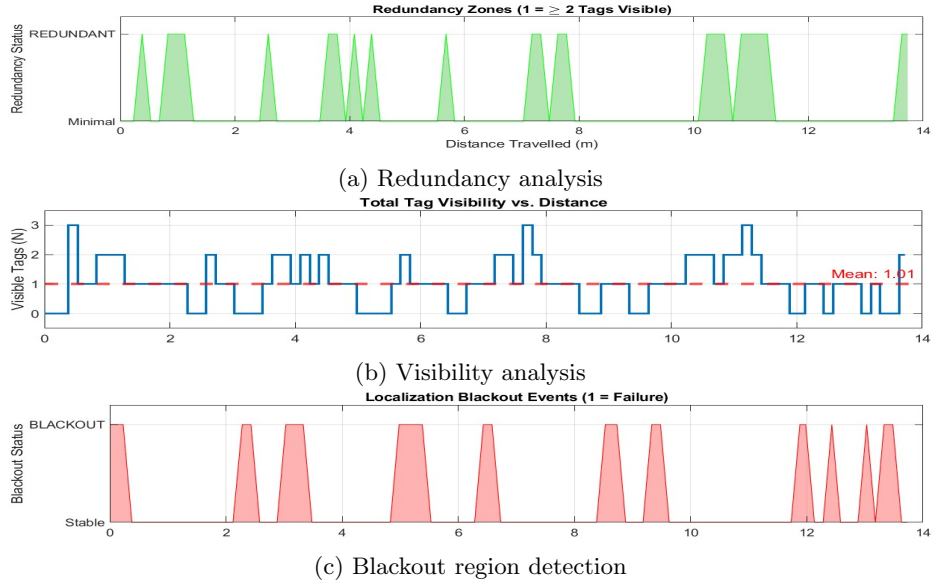


Fig. 6: Tag visibility analysis in simulation environment

Table 4: Performance validation in simulation environment

Metric	Value
Navigation Distance	13.73 m
Mean Tags Visibility	1.011
Max Tags Visible	3
Localization Reliability (Path with ≥ 1 Tag)	94.58%
Blackout Rate (Failure) (Path with 0 Tags)	6.32%
Redundancy Rate (Optimal) (Path with ≥ 2 Tags)	24.21%

4.2 Performance validation in real-time environment

The proposed work is also evaluated in a real-world environment using a scaled-down model as in Figure 1a. Kinematic constraints and dimensions of the Yahboom Jetson robot used in this work are considered. A feasibility analysis results in a 10 cm ArUco tag for the given environment and camera intrinsic parameters. Experimental analysis results in a tag availability of 92% (from Table 5), which is identical to the simulation (Tag availability: 94%). Observations also shows that the alternating placement strategy provides better coverage-to-tag ratio compared to other constraints like single side and randomized placement. Prioritizing structurally critical areas like junctions and task areas results in reliable navigation. Notably, the proposed work also achieve a 100% visibility in the junction enables the robot to localize accurately and eliminating the loss of markers during a change in robot heading. Thus, the proposed technique directly addresses a major challenge in indoor robot navigation and also resolves visual ambiguity in repetitive environments while limiting the infrastructure changes.

Table 5: Performance validation in real-time environment

Metric	Value
Total Frames	2666
Blackout Percentage	8.18%
Availability Percentage	91.82%

5 Conclusion

Visual landmarks like ArUco markers provide reliable beacons for robot navigation in a structurally repetitive indoor environment. A hierarchical three-stage optimization algorithm is proposed to place the ArUco markers, enabling robot navigation. Inference generated from the manual placement provides insight into tag placement. Visibility awareness and structure-specific deployment of tags are considered in this study. A greedy search algorithm is used to optimize the tag placement. Experimental analysis results in 94.2 % of tag availability. Validation in the simulation and real-world environment also results in a mean visibility of 1.011 tags per frame.

Development of an end-to-end automated strategy for automated marker placement using machine learning techniques will be the extension of the proposed work. Integration of sensor fusion techniques to combine visual markers information with IMU and wheel odometry will be investigated in the future.

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